



AquaGrid

An app that works out the smartest time to make drinking water on a small island — so the island wastes less fuel and never runs dry.

TEAM

BitHeads

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DreamHacks 2026 · Track 2 — AI, Automation & Logic

01

The problem

Why an island can't just turn on a tap

RISHIK

Small islands have to **make** their water

Rain fills the tanks. When the rain stops, a machine turns seawater into drinking water.

That machine is the **hungriest thing on the island** for electricity.

By day that power is free — it's sunshine. At night it's a diesel generator, and the diesel comes **by boat every few weeks**.

The tank is basically a **battery for water**. You can fill it whenever you like — so you should fill it when power is free.

Midday

sunshine — free

2 a.m.

diesel — costs money

The tank

holds about 2 days

But nobody chooses. A timer does.

Most islands run the machine on a **timer set to the middle of the night**, because that's when electricity is cheap on the mainland.

On a solar island that's backwards — the middle of the night is exactly when there is **no sun**.

This is a real problem. In **2011 Tuvalu ran down to a few days of drinking water** and had to declare a state of emergency.

Their machine wasn't too small. The problem was that **nobody could see what was coming**.

0 2

The solution

Teaching it to see what's coming

NEEKIN

So we taught it to see what's coming

01

Predict

How much water the island will use, 2 days ahead

02

Decide

The best hours to run the machine

03

Explain

Says why, in plain English

04

Detect

Spots problems early

We trained an AI on **2 years of water use**. Then we tested it on days it had **never seen before** — real future days, not a shuffled deck.

45 litres

how far off it is, per hour — out of about 940

0.987

accuracy score, where 1.0 is perfect

89%

better than just guessing the average

It uses **a fifth** of the fuel

HOW IT'S RUN	DIESEL USED	COST	FUEL LASTS	RESULT
AquaGrid	15 L	\$25	39 days	fine
Night timer	70 L	\$112	8.6 days	runs out
Fill when low	79 L	\$127	7.6 days	runs out

Here's the part that really matters. The island has **300 litres of diesel** and the boat is **12 days away**.

The other two ways **run out of fuel before the boat arrives**. Ours doesn't.

03

The architecture

All of it, running on a phone

SRIHAN

The whole thing runs on the phone

THE APP	React Native	what you see and tap — we drew every chart ourselves
THE BRAIN	RandomForest AI	we trained it on a laptop, then packed it inside the app
THE WEATHER	Open-Meteo	the phone grabs its own sun and rain forecast — free, no sign-up
THE VOICE	Groq • GPT-OSS 120B	an open AI model that explains things in plain English

There is **no server anywhere**. An app that needs the internet to decide when to make water would stop working **exactly when an island gets cut off** — which is when you need it most.

We put the whole AI **inside** the app

Normally an AI lives on a big computer and your phone asks it questions over the internet. We did the maths **ahead of time** for every situation the AI could ever face, and packed all the answers into the app.

15,120

answers – the whole AI, not a sample

0.011_{ms}

time to predict a whole day, on the phone

Any island

works for places the laptop never saw

The RandomForest

We trained this. It makes

every

decision.

The language model

Only writes the words. Delete it and

every number stays the same

.

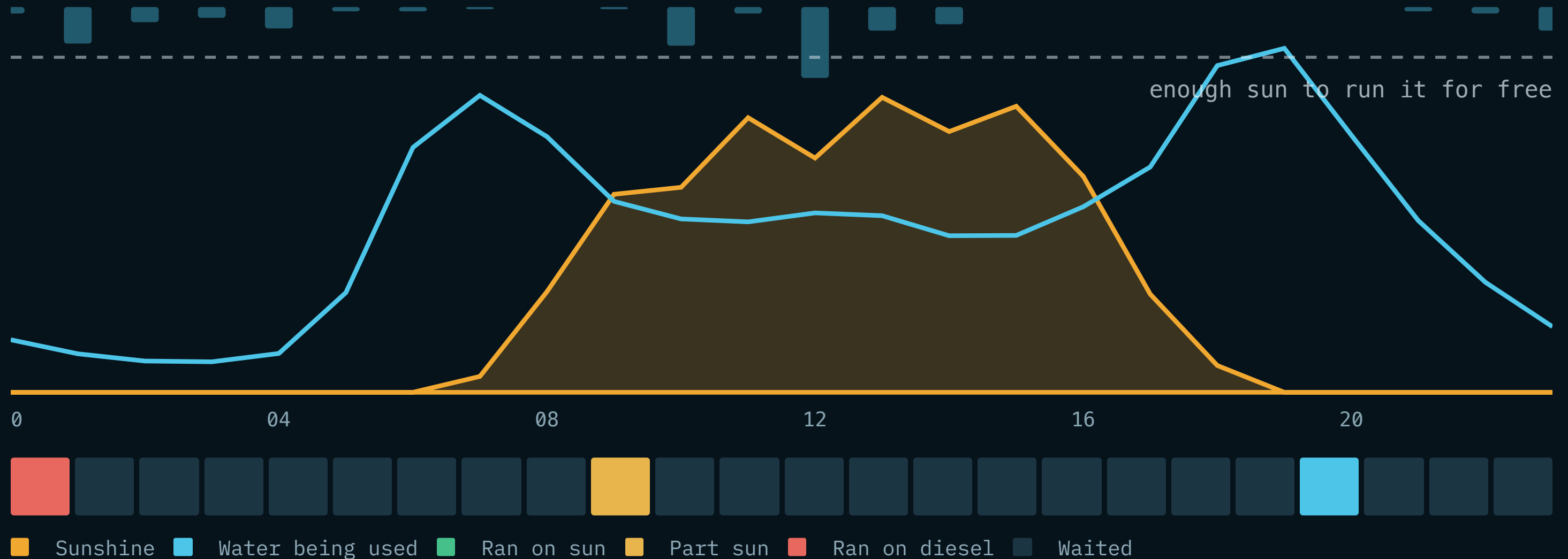
0 4

The features

One real day, and the problems it catches

EDNIT

One real day



Orange above the dotted line means the sun alone can run the machine. The strip underneath is what it actually chose to do, hour by hour.

It also catches problems early

It knows how fast the tank **should** empty. So when the real tank drops faster than that, something is wrong — and it says so.

Burst pipe

found in 2 hours

Dirty solar panels

losing 13% of the sunshine

Worn-out filters

need cleaning in 40 days

Running low on fuel

before the boat comes

It saves the rain too. Rain off the rooftops gave **24%** of all the water — so if rain is coming, it waits instead of wasting fuel.

And it stores sunshine. A battery carries midday sun into the evening, when everyone gets home and uses the most water.

05

The demo

Let's actually show you

EVERYONE

Let's show you

- | | | |
|-----------------------------------|---------------------------------|--|
| 1 What it's doing right now — and | why , in a full sentence | 2 The day ahead: green = free sun, blue = stored sun, red = diesel |
| 3 How much fuel it saves | | 4 Press "burst pipe" — watch the alarm appear |

Nobody would have spotted that leak for hours. The app found it in two.

And then

- | | |
|--|--|
| 5 Press “cyclone” — it stops saving fuel and fills the tank | 6 Switch to a different island — it works everything out again |
| 7 Ask it a question. Then ask “why?” | 8 Turn on aeroplane mode — everything still works |

Because on a real island, the internet is the first thing to go.

06

Code & website

How it's built, and how to check us

GOWTHAM

How it's built

755

lines of code make all the decisions

One
command

tests it all – no phone needed

Free

the weather data costs nothing and
needs no sign-up

ml/	Teaches the AI	2 years of water use → a trained model
logic/	Makes the decisions	when to run, and spotting problems
lib/	Talks to the world	weather, location, the language model
components/	Draws the screen	18 pieces, every chart hand-drawn

SEE EVERYTHING – CODE, DATA, WORKINGS

github.com/sbongu1a/aquagrid

Everything is written down

Website

the charts are made from real data,
not screenshots

35_{pages}

a handbook explaining every file

All open

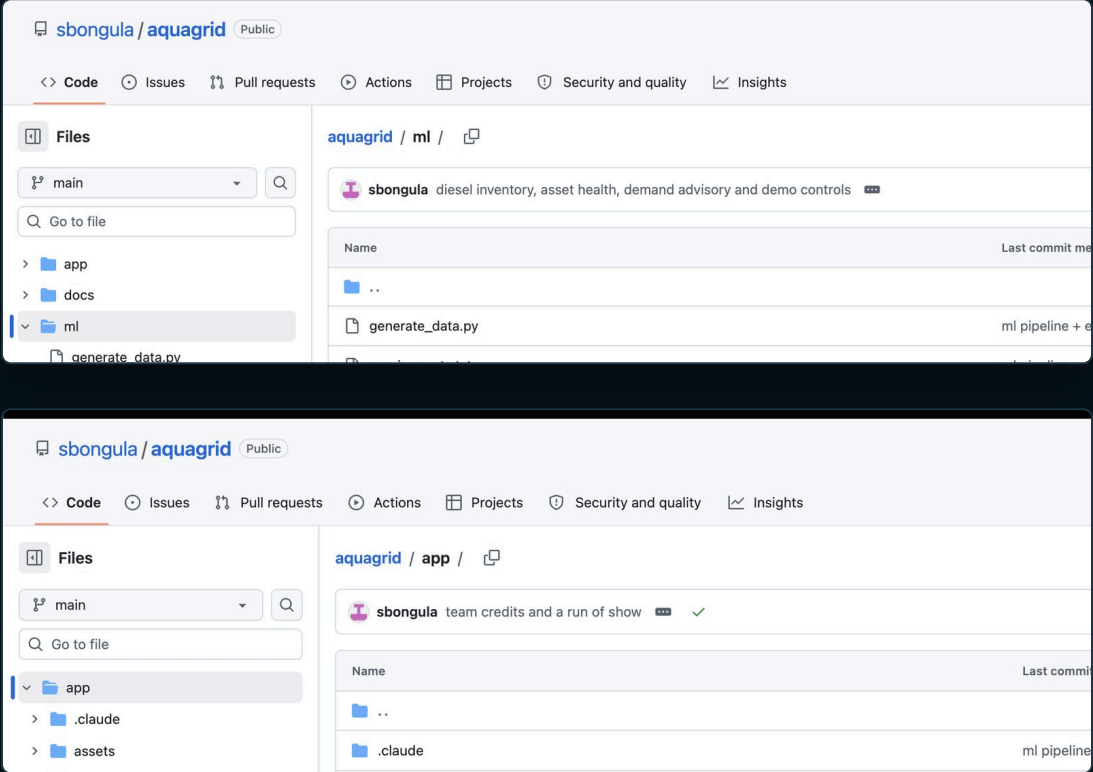
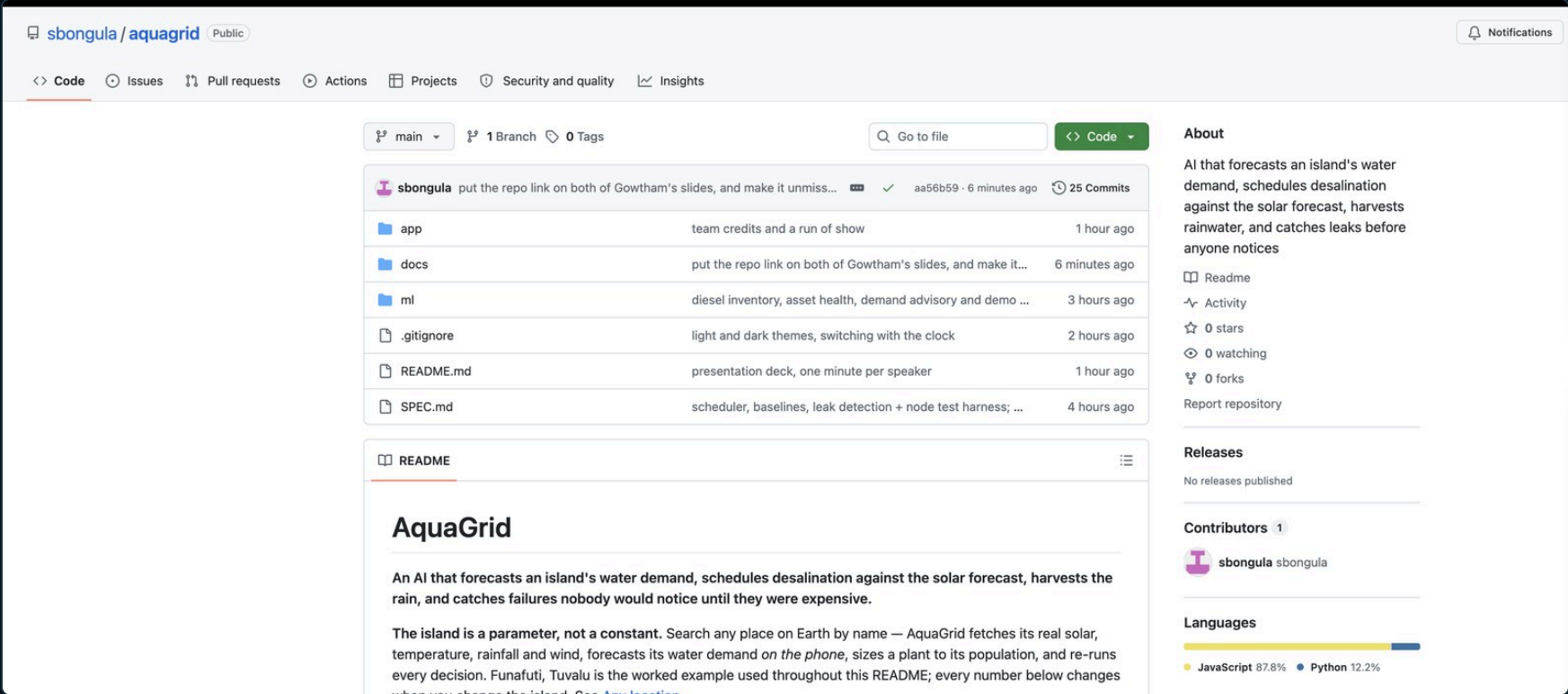
anyone can read it and check our
numbers

We didn't just say it saves fuel. **We ran three different ways of doing it** — ours, the night timer, and fill-when-low — on the same weather, and compared them.

CHECK OUR NUMBERS YOURSELF

github.com/sbongu1a/aquagrid

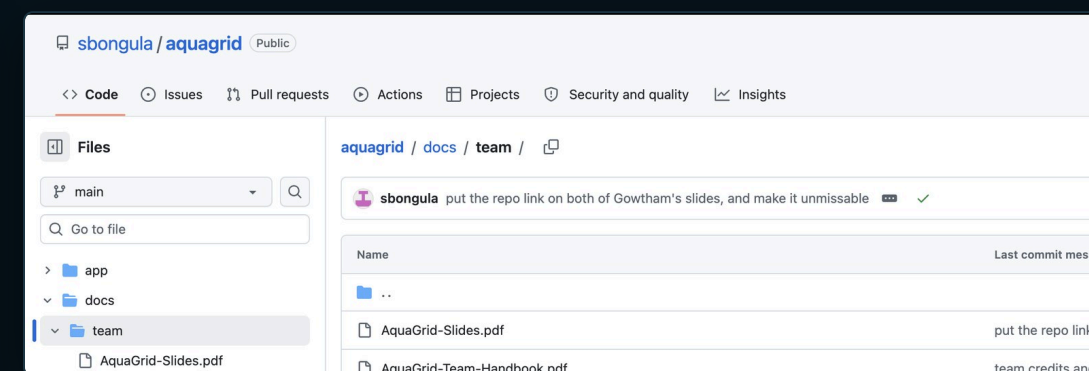
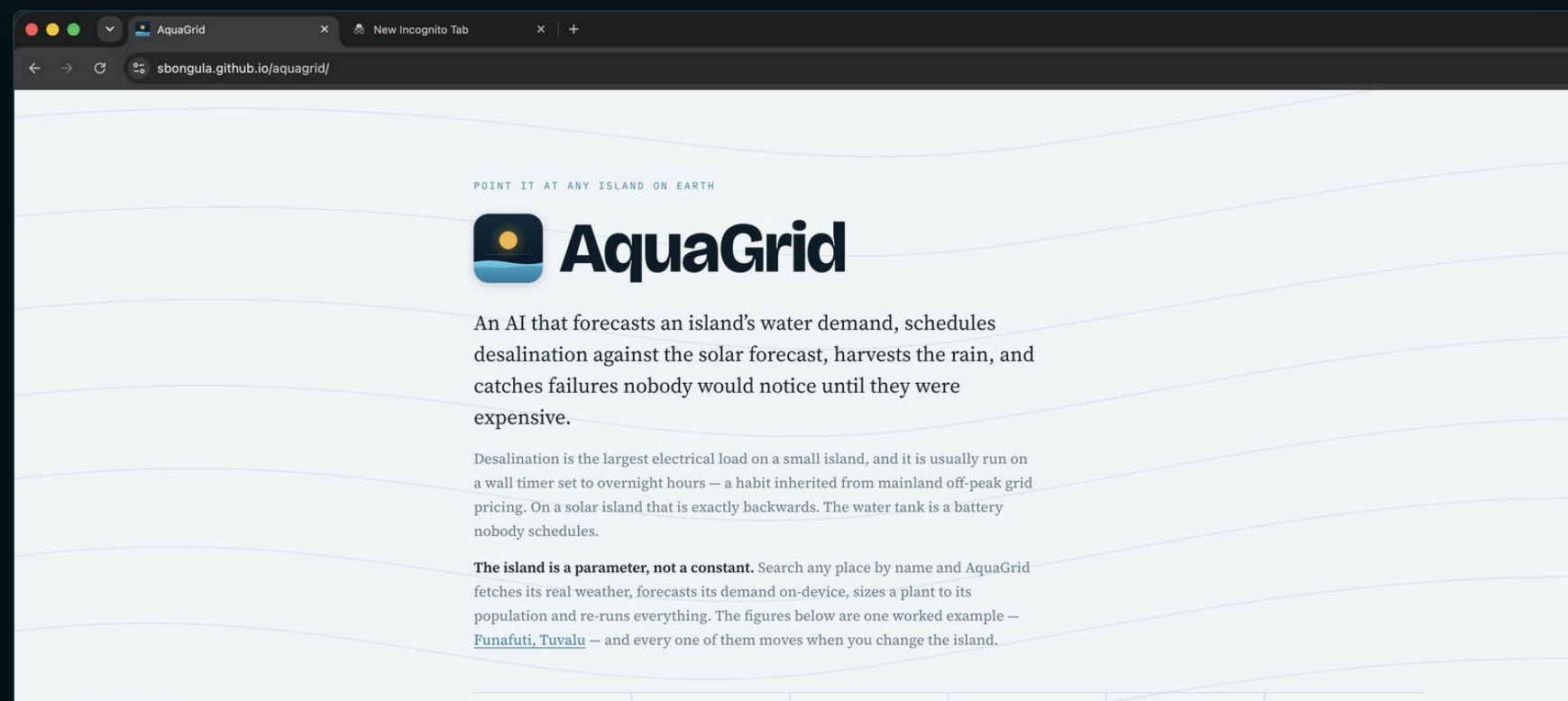
It's all public — go and check



EVERY LINE, EVERY NUMBER

github.com/sbongula/aquagrid

And a website that explains all of it



Charts built from real data

not screenshots — so they can't go out of date

READ IT YOURSELF

sbongula.github.io/aquagrid



Every number here is one we measured.

We tested the AI on days it had never seen. We gave the leak detector a real leak and made it find it. And when we tried Reykjavík — where there's barely any sun — it only saved 23%. We left that in, because a smaller honest number is worth more than a big one we can't back up.

github.com/sbongula/aquagrid

sbongula.github.io/aquagrid